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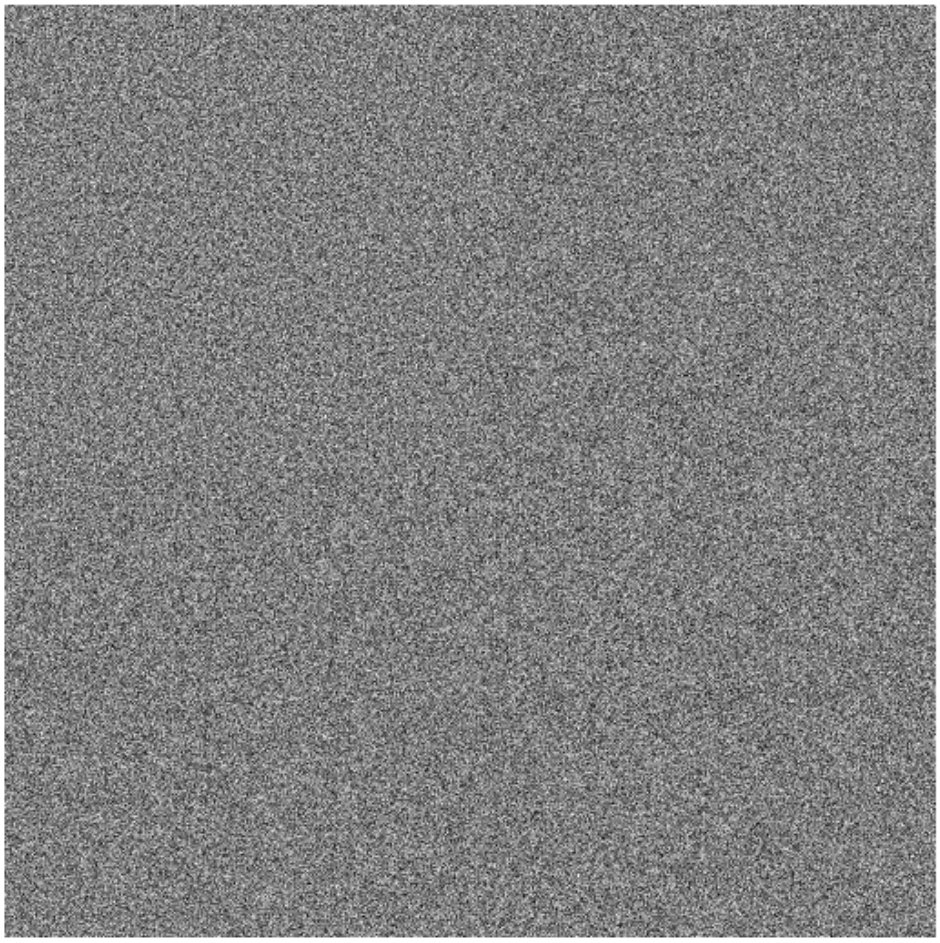
Load Data

```
%Loading the hidden message for processing
clear;
clc;
close all;
load('hiddenmsg.mat');
C =double(hiddenmsg);
N=size(C, 1);
```

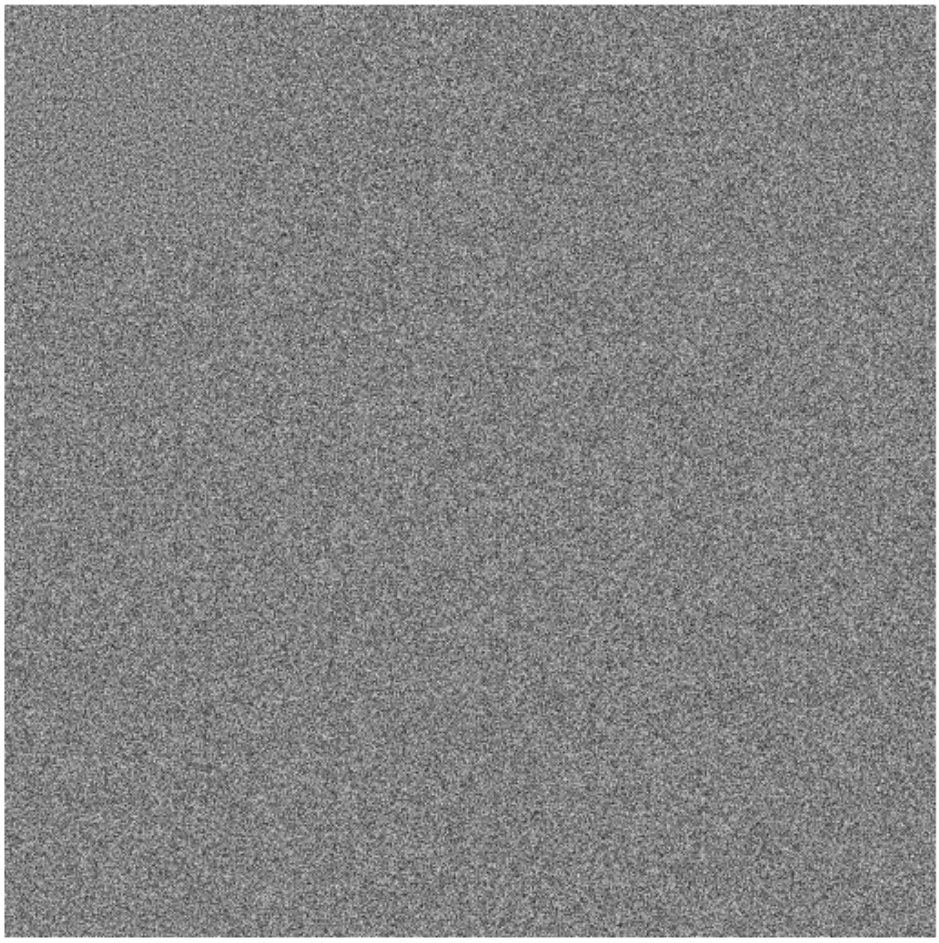
Iterative Haar Transform

```
%Applying the haar transform in 4 stages
for k= 1:4
    n = N / 2^(k-1);
    P= iterate_haar(n);
    C(1:n,1:n) = P *C(1:n, 1:n)*P';
    figure;
    imshow(C, []);
    title(['Haar Transform Stage ',num2str(k)]);
end
```

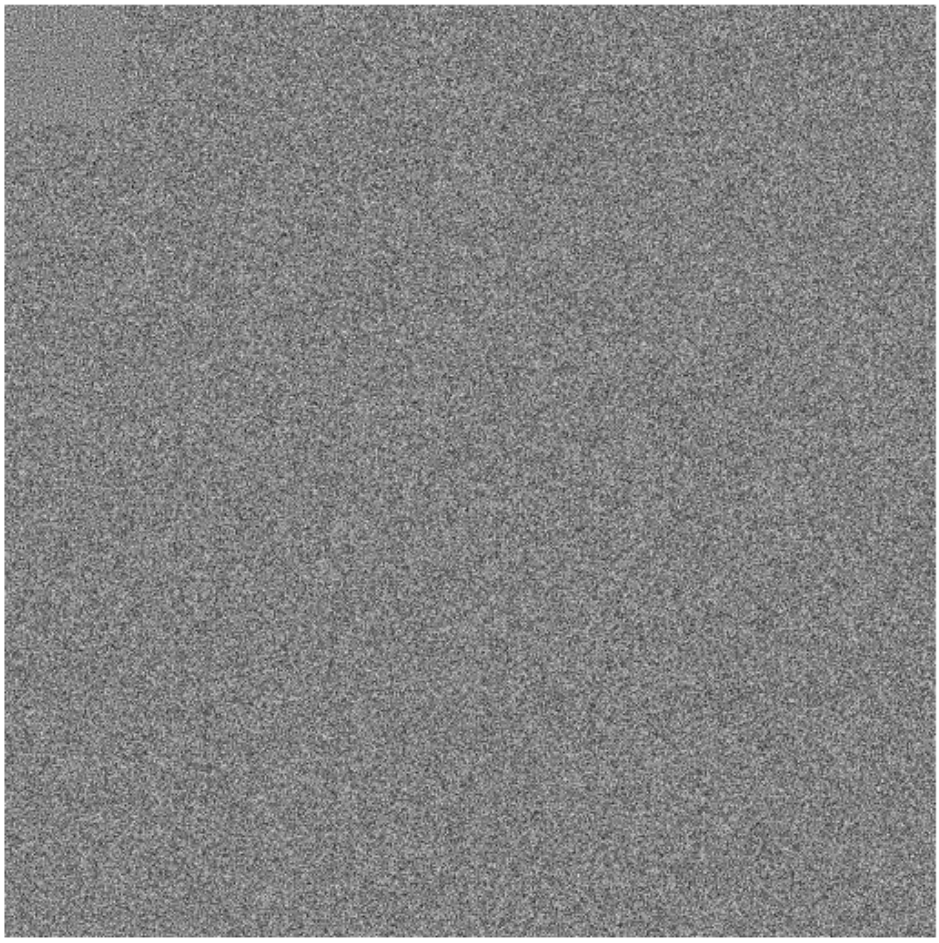
Haar Transform Stage 1

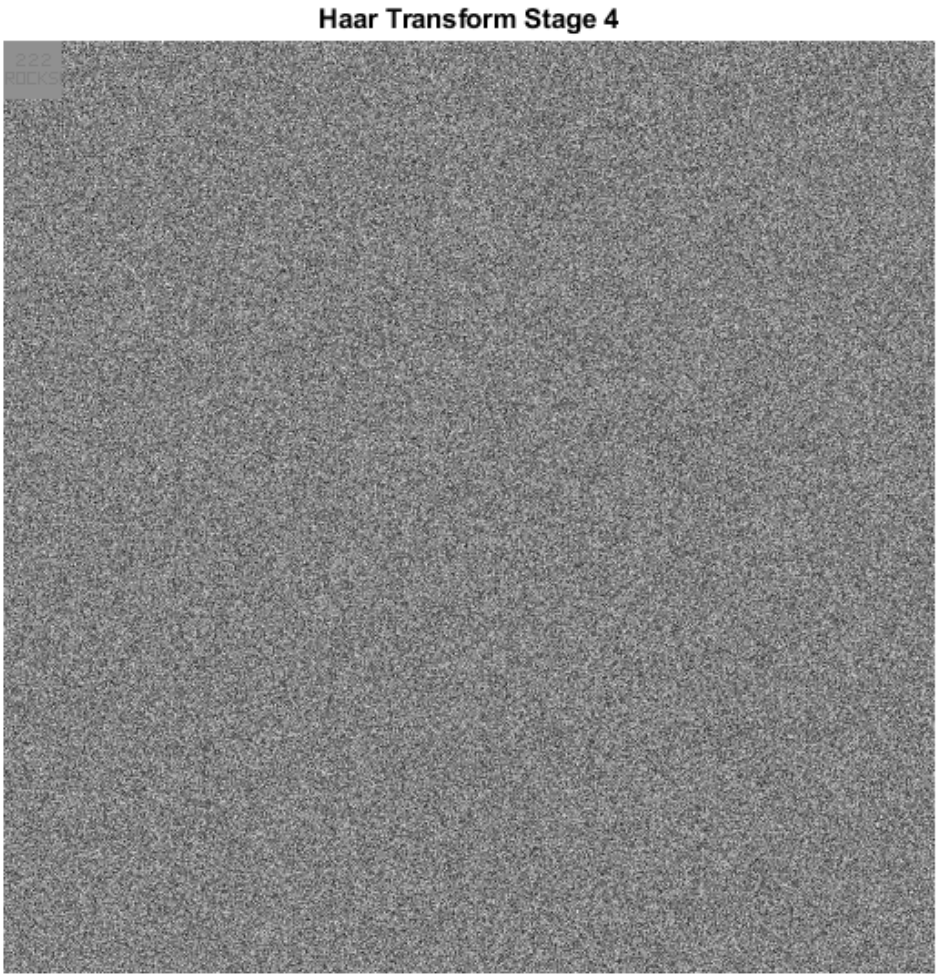


Haar Transform Stage 2



Haar Transform Stage 3





Determine L

Examining the results of the iterative Haar wavelet transform. Viewing the top left block of the transformed matrix.

```
%At the fourth stage, the block is close to being uniform.
% This is the separation between the message and noise component.
% Since the image has 10 wavelet scales, and four stages are applied to it.
% the resulting separation level is L = 10 - 4 = 6.
% Therefore, the determined value of L is 6.
L = 6;
disp(['L = ', num2str(L)]);
```

```
L = 6
```

Remove Noise Coefficients

Removing the noise coefficients while keeping the message information.

```
n = N / 2^(10-L);
Cclean = zeros(size(C));
Cclean(1:n, 1:n) = C(1:n, 1:n);
```

Inverse Haar Transform

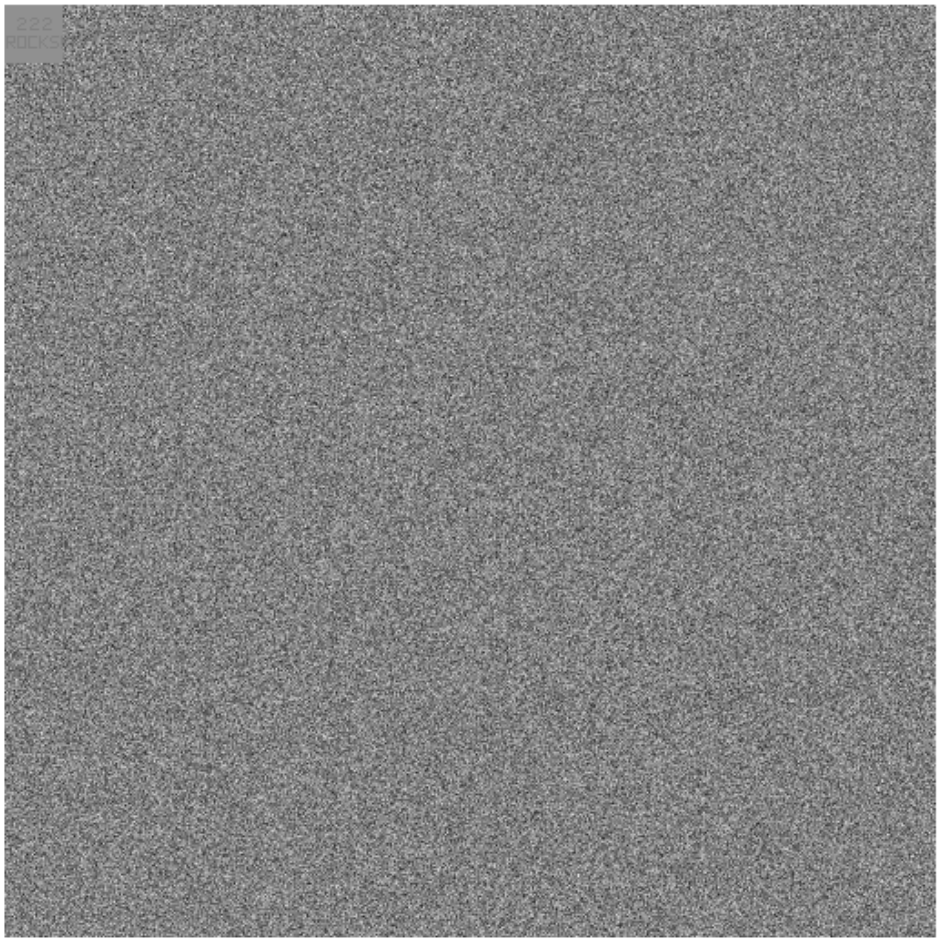
Reversing the Haar transform to reconstruct the original image.

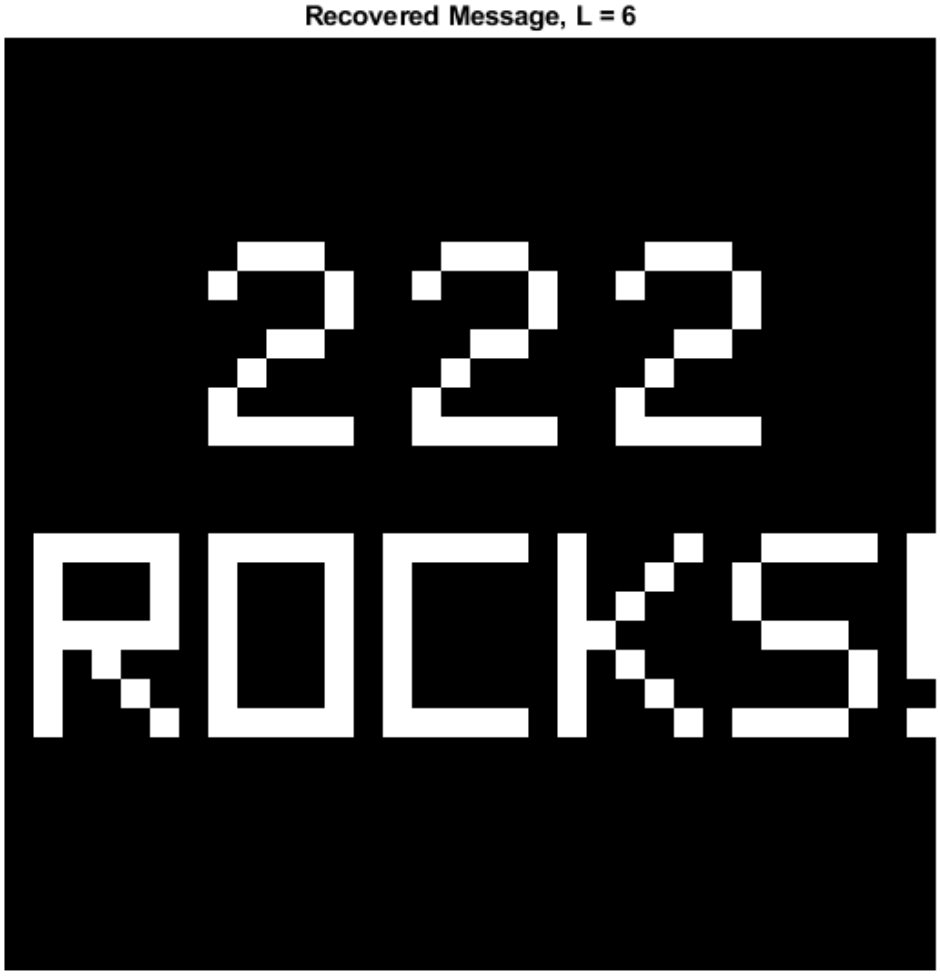
```
for k = 4:-1:1
    n = N / 2^(k-1);
    P = iterate_haar(n);
    Cclean(1:n, 1:n) = P'*Cclean(1:n, 1:n) * P;
end
```

Display Recovered Message

```
%Displaying the recovered image after the transformation
figure;
imshow(-Cclean, [])
title(['Recovered Message, L = ',num2str(L)]);
```

Haar Transform Stage 4





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